Lecture 3

Which momentum? Self-adjointness is a choice

Objectives. *The rule $-i\hbar\psi'$ becomes different operators on different domains. On the circle it is self-adjoint. In the box the Dirichlet momentum is only symmetric, and its self-adjoint extensions are the momenta P_θ. We compare their readings and relate finite mean energy to the Dirichlet domain.*

- Use the boundary form to test symmetry.
- Compare the P_0 and P_π readings of one box state.
- Explain why no self-adjoint extension of P_D squares to the box energy.

Lecture 3, first part

The momentum on a circle and the momentum with walls

3.1 From the ring to the circle

Take the ring of *Quantum Mechanics I*, Lecture 8: N sites, the shift T with $Te_x = e_{x+1}$, and the Hamiltonian $c\mathbb{1} - \kappa(T + T^\dagger)$, whose eigenvectors are the Fourier modes f_k, with the levels $c - 2\kappa\cos(2\pi k/N)$. The circle's functions f_k below, sampled at the sites, are these modes up to normalisation, so they share the letter. As for the chain of § 1.1, put $c = 2\kappa$, the spacing $a = L/N$ and $\kappa = \hbar^2/2ma^2$. The ring now has the circumference L, and its levels are

$$4\kappa \sin^2 \frac{\pi k}{N} = \frac{2\hbar^2 N^2}{mL^2} \sin^2 \frac{\pi k}{N} \longrightarrow \frac{p_k^2}{2m}, \qquad p_k = \frac{2\pi\hbar k}{L},$$

at each fixed integer k as $N \to \infty$, from below. In units of the box's $E_1 = \hbar^2\pi^2/2mL^2$ these are $p_k^2/2m = 4k^2E_1$: the levels 0, then $4E_1$ twice ($k = \pm 1$), $16E_1$ twice, and so on. They are not the box's $E_1, 4E_1, 9E_1, \ldots$. The ring's limit is a particle on a circle.

A particle on a circle of circumference L has a position x in $[0, L)$, where $x = L$ is the point $x = 0$ again. Its wavefunctions form the same space $L^2(0, L)$, with the same inner product, and the functions

$$f_k(x) = \frac{e^{2\pi i k x/L}}{\sqrt{L}} \qquad (k \in \mathbb{Z})$$

form a complete orthonormal basis. This is the Fourier basis of *Linear Algebra*, Ex. 13.25, rescaled, and a standing fact. Write $c_k = \langle f_k, \psi \rangle$.

Definition 3.1 (The momentum on a circle)**.** The momentum P on the circle has the

domain and the action

$$D(P) = \Big\{\psi \in L^2(0,L) : \sum_{k\in\mathbb{Z}} k^2\,|c_k|^2 < \infty\Big\}, \qquad P\psi = \sum_{k\in\mathbb{Z}} p_k c_k f_k.$$

Proposition 3.2 (The momentum on a circle)**.** *The momentum P is densely defined and self-adjoint, and its eigenvalues are exactly the p_k, and the eigenvectors for p_k are the nonzero multiples of f_k. Every continuously differentiable ψ on $[0,L]$ with $\psi(L) = \psi(0)$ lies in $D(P)$, and $P\psi = -\mathrm{i}\hbar\psi'$.*

Proof. The steps of the proof of Theorem 1.8 on density, symmetry, the adjoint and the eigenvalues apply word for word, with f_k, p_k and $k \in \mathbb{Z}$ in place of u_n, E_n and $n \geq 1$, since p_k^2 is a multiple of k^2 and the p_k are distinct and real. For the rule, let ψ be continuously differentiable with $\psi(L) = \psi(0)$. Integrating by parts,

$$\langle f_k, -\mathrm{i}\hbar\psi'\rangle = -\mathrm{i}\hbar\left[\overline{f_k}\psi\right]_0^L + \left\langle -\mathrm{i}\hbar f_k', \psi\right\rangle.$$

The bracket is $-\mathrm{i}\hbar(\psi(L) - \psi(0))/\sqrt{L} = 0$, and $-\mathrm{i}\hbar f_k' = p_k f_k$, so the coefficients of $-\mathrm{i}\hbar\psi'$ are $p_k c_k$. By Parseval's identity, $\sum_k p_k^2\,|c_k|^2 = \hbar^2\,\|\psi'\|^2 < \infty$, so $\psi \in D(P)$ and $P\psi = -\mathrm{i}\hbar\psi'$. ∎

We take as the energy of the particle on the circle the diagonal operator with the entries $p_k^2/2m$, which is $P^2/2m$. Its entries are the limits of the ring's levels.

Self-check 3.1

1. List the three lowest levels of the circle and of the box, with their multiplicities, in units of E_1.

3.2 The class $\mathcal{A}$ and the boundary form

We now study the momentum in the box. The rule $-\mathrm{i}\hbar\psi'$ needs a derivative and values at the walls, and an element of $L^2(0,L)$ has neither (§ 1.2). Continuous derivatives are too few. The tent $\tau(x) = \min(x, L-x)$ has a corner at $L/2$, and it lies in the domain of the circle's momentum, since its coefficients c_k vanish for even $k \neq 0$ and are $-L^{3/2}/\pi^2k^2$ for odd k, so that $\sum_k k^2\,|c_k|^2 < \infty$. What is needed is a derivative that is square-integrable.

Definition 3.3 (The class $\mathcal{A}$)**.** Let $\mathcal{A}$ be the set of $\psi \in L^2(0,L)$ whose class contains a function of the form

$$\psi(x) = a + \int_0^x g(t)\,\mathrm{d}t \qquad (0 \leq x \leq L),$$

with $a \in \mathbb{C}$ and $g \in L^2(0,L)$. We write $\psi' = g$.

They are the absolutely continuous functions with a square-integrable derivative. The following properties make them usable.

1. *Values.* By Cauchy–Schwarz, $|\psi(y) - \psi(x)| \leq \sqrt{y-x}\;\|g\|$ for $x \leq y$, so ψ is continuous. Two continuous functions that differ at a point differ on an interval, so a class contains at most one continuous function. Hence $\psi(0) = a$ and $\psi(L)$ are well defined.
2. *The derivative.* Suppose that ψ has two such forms, with g and $\tilde{g}$. Then both constants are $\psi(0)$, and $d = g - \tilde{g}$ has $\int_0^x d(t)\,\mathrm{d}t = 0$ for every x. So d is orthogonal to the indicator of every interval $[0,x]$, hence to every step function. Step functions approximate

continuous functions uniformly, and continuous functions are dense in $L^2(0,L)$, since the smooth functions are (the standing fact S4). So $d = 0$, and the derivative ψ' is determined by ψ.

3. *Integration by parts.* For $\varphi, \psi \in \mathcal{A}$,

$$\int_0^L \overline{\varphi}\,\psi'\,\mathrm{d}x = \big[\overline{\varphi}\psi\big]_0^L - \int_0^L \overline{\varphi'}\,\psi\,\mathrm{d}x. \tag{3.1}$$

   For continuous φ' and ψ' this is calculus. In general, take continuous $g_j \to \psi'$ and $h_j \to \varphi'$ in $L^2(0,L)$, and their primitives $\psi_j = \psi(0) + \int_0^x g_j$ and $\varphi_j = \varphi(0) + \int_0^x h_j$. By the first fact, $\psi_j \to \psi$ and $\varphi_j \to \varphi$ uniformly, and every term of (3.1) converges. The same holds on every interval $[0,y]$ in place of $[0,L]$.

4. *The product rule.* Let $\psi \in \mathcal{A}$ and let b be continuously differentiable on $[0,L]$. Then $\overline{b} \in \mathcal{A}$, and (3.1) on $[0,y]$, with $\varphi = \overline{b}$, gives $b(y)\psi(y) = b(0)\psi(0) + \int_0^y (b'\psi + b\psi')\,\mathrm{d}t$. So $b\psi \in \mathcal{A}$, with $(b\psi)' = b'\psi + b\psi'$.

For example, the tent lies in $\mathcal{A}$: $\tau(x) = \int_0^x g(t)\,\mathrm{d}t$, with $g = 1$ on $[0,L/2]$ and $g = -1$ on $(L/2,L]$. A function with a jump does not, since by the first fact every element of $\mathcal{A}$ has a continuous function in its class. Nor does $\sqrt{x}$, although it is continuous. On every $[\epsilon,L]$ its derivative is $1/2\sqrt{x}$, and by the second fact, applied on $[\epsilon,L]$, so is any g in a form of $\sqrt{x}$. But $\int_\epsilon^L \mathrm{d}x/4x$ grows without bound as $\epsilon \to 0$, so no such g is square-integrable.

A derivative in $\mathcal{A}$ may still be unbounded. The function $x^{3/4}$ lies in $\mathcal{A}$, with $g = \frac{3}{4}x^{-1/4}$ and $\int_0^L g^2\,\mathrm{d}x = \frac{9}{8}\sqrt{L}$. Among the primitives $a + \int_0^x g$, the integral of g^2 decides membership, not the size of g.

Proposition 3.4 (The boundary form). *For $\varphi, \psi \in \mathcal{A}$,*

$$\langle \varphi, -\mathrm{i}\hbar\psi' \rangle - \langle -\mathrm{i}\hbar\varphi', \psi \rangle = -\mathrm{i}\hbar\big(\overline{\varphi(L)}\psi(L) - \overline{\varphi(0)}\psi(0)\big). \tag{3.2}$$

Proof. Multiply (3.1) by $-\mathrm{i}\hbar$, and use $\overline{-\mathrm{i}\hbar\varphi'} = \mathrm{i}\hbar\overline{\varphi'}$. ∎

The rule on all of $\mathcal{A}$ is not symmetric: for $\psi(x) = x$ the number $\langle \psi, -\mathrm{i}\hbar\psi' \rangle = -\mathrm{i}\hbar L^2/2$ is not real. The walls suggest the boundary condition that removes the right side.

Definition 3.5 (The momentum with walls). The *Dirichlet momentum* P_D has the domain $D(P_D) = \{\psi \in \mathcal{A} : \ \psi(0) = \psi(L) = 0\}$ and the action $P_D\psi = -\mathrm{i}\hbar\psi'$.

Proposition 3.6 (The momentum with walls is symmetric). *The Dirichlet momentum P_D is densely defined and symmetric.*

Proof. Each u_n lies in $D(P_D)$, so the finite combinations of the u_n do, and they are dense. For $\varphi, \psi \in D(P_D)$, the right side of (3.2) vanishes. ∎

The momentum P_D is what physics texts call the Hermitian momentum of the box.

Figure 3.1 draws the tent, the function $x^{3/4}$, the function $\sqrt{x}$ and the indicator of $[0,1/2]$, on $[0,L]$ with $L = 1$, and it marks which of them lie in $\mathcal{A}$ and in $D(P_D)$.

Self-check 3.2

1. Is $\psi(x) = x(L-x)$ in $D(P_D)$? Compute $\langle \psi, P_D\psi \rangle$.
2. Which of these lie in $\mathcal{A}$: the indicator of $[L/4,L/2]$, the function $x^{2/3}$, the function $x^{2/3}(L-x)$? Which lie in $D(P_D)$?

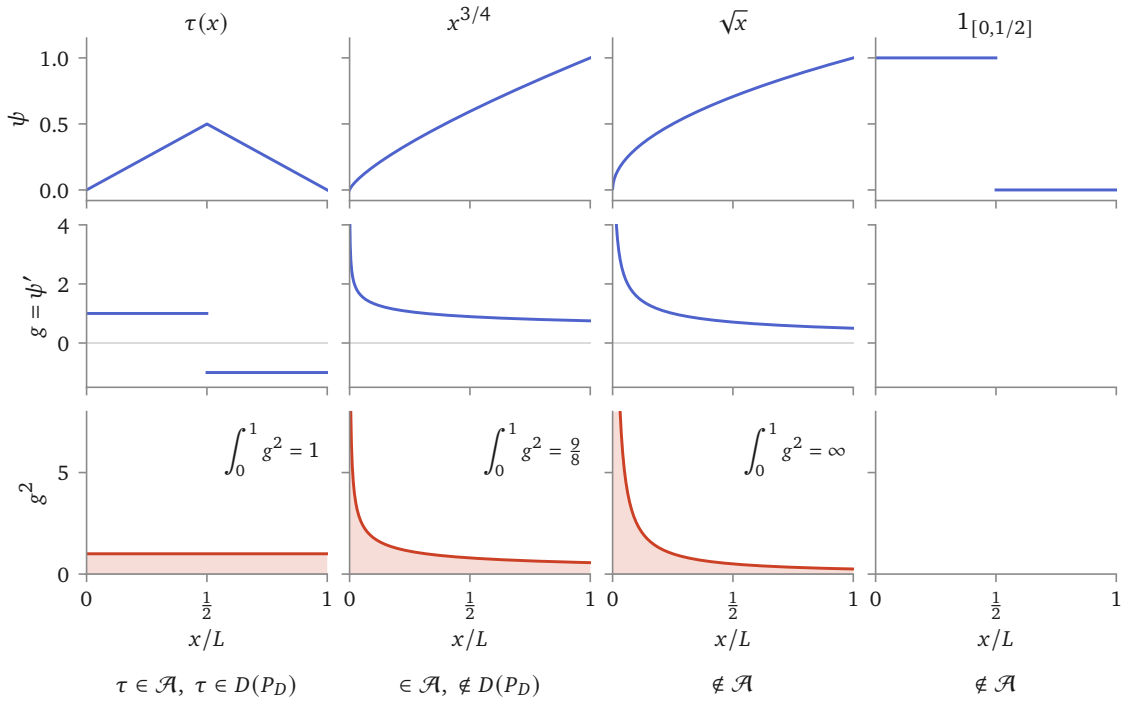


Figure 3.1: Four functions on $[0, L]$, with $L = 1$. Top row: the functions, with a gap at the jump of the indicator. Middle row: the function g whose primitive $a + \int_0^x g$ is the curve above, where there is one. The indicator has none, since such a primitive is continuous. Off its jump its derivative is 0, and the primitive of 0 is a constant. Bottom row: g^2, whose integral over $(0, L)$ decides membership of $\mathcal{A}$. The last line says which functions lie in $\mathcal{A}$ and in $D(P_D)$.

3.3 The adjoint of the momentum with walls

Recall that $A \subseteq B$ means that B extends A (Definition 0.4), and that adjoints reverse inclusions (Proposition 0.5).

Proposition 3.7 (A large adjoint). *Every $\varphi \in \mathcal{A}$ lies in $D(P_D^\dagger)$, with $P_D^\dagger \varphi = -i\hbar\varphi'$. Every complex number z is an eigenvalue of $P_D^\dagger$, with the eigenvector $e^{izx/\hbar}$. The operator P_D has no eigenvalue, and it is not self-adjoint.*

Proof. Let $\varphi \in \mathcal{A}$ and $\psi \in D(P_D)$. Since $\psi(0) = \psi(L) = 0$, (3.2) gives $\langle \varphi, P_D\psi\rangle = \langle -i\hbar\varphi', \psi\rangle$. By Definition 0.2, with $\chi = -i\hbar\varphi'$, the vector φ lies in $D(P_D^\dagger)$. The function $e^{izx/\hbar}$ lies in $\mathcal{A}$, and $-i\hbar(e^{izx/\hbar})' = ze^{izx/\hbar}$. If $-i\hbar\psi' = z\psi$ for some $\psi \in \mathcal{A}$, then by the product rule $(e^{-izx/\hbar}\psi)' = e^{-izx/\hbar}(\psi' - iz\psi/\hbar) = 0$, so $e^{-izx/\hbar}\psi$ is the constant $\psi(0)$, and $\psi = \psi(0)\, e^{izx/\hbar}$. In $D(P_D)$, $\psi(0) = 0$ and $\psi = 0$. Finally, the constant 1 lies in $D(P_D^\dagger)$ and not in $D(P_D)$. ∎

The domain of the adjoint contains nothing else: it is exactly the class $\mathcal{A}$.

Theorem 3.8 (The adjoint). *$D(P_D^\dagger) = \mathcal{A}$. So $P_D^\dagger$ is the rule $-i\hbar\varphi'$ with no boundary condition.*

Proof. We prove both inclusions. First, Proposition 3.7 gives $\mathcal{A} \subseteq D(P_D^\dagger)$, with the action $P_D^\dagger\varphi = -i\hbar\varphi'$.

Conversely, let $\varphi \in D(P_D^\dagger)$, and write $\chi = P_D^\dagger \varphi$. By Definition 0.2, $\langle \varphi, -i\hbar\psi' \rangle = \langle \chi, \psi \rangle$ for every $\psi \in D(P_D)$. We move both sides onto ψ'. Since the inner product is antilinear in its first (left) slot, $\langle \varphi, -i\hbar\psi' \rangle = \langle i\hbar\varphi, \psi' \rangle$. Put $G(x) = \int_0^x \chi(t)\,dt$, so that $G \in \mathcal{A}$ and $G' = \chi$. Since ψ vanishes at the walls, integration by parts (3.1) gives $\langle \chi, \psi \rangle = -\langle G, \psi' \rangle$. Hence

$$\langle i\hbar\varphi + G, \psi' \rangle = 0 \quad \text{for every } \psi \in D(P_D).$$

Every square-integrable g with $\int_0^L g = 0$ is such a ψ', and this fixes φ. Let c be the average of $h = i\hbar\varphi + G$ over $(0, L)$, and put $g = h - c$. Then $\int_0^L g = 0$, so $\psi(x) = \int_0^x g(t)\,dt$ lies in $D(P_D)$, with $\psi' = g$. The display gives $\langle h, g \rangle = 0$, and $\langle c, g \rangle = \bar{c} \int_0^L g = 0$. So $\|g\|^2 = \langle h - c, g \rangle = 0$, and $h = c$. Hence $\varphi = (c - G)/i\hbar$, which lies in $\mathcal{A}$, and $-i\hbar\varphi' = G' = \chi$. ∎

So P_D is too small, and its adjoint is too large: P_D has no eigenvalue, and $P_D^\dagger$ has every complex number as one. The symmetric operator P_D is not an observable.

Self-check 3.3

1. Give an eigenvector of $P_D^\dagger$ with the eigenvalue $i\hbar/L$. Why is it not an eigenvector of P_D?

Lecture 3, second part

The self-adjoint momenta in a box and the box's energy

3.4 The momenta P_θ

The first part showed that the momentum with walls P_D is symmetric and not self-adjoint, and that its adjoint is the rule $-i\hbar\psi'$ on all of $\mathcal{A}$. In this part we find every self-adjoint extension of P_D, compare the readings of two of them on one state, and realise them on a ring with a magnetic flux. We then relate the momentum with walls to the energy of the box.

A self-adjoint momentum P in the box with $P_D \subseteq P$ satisfies $P \subseteq P_D^\dagger$, since adjoints reverse inclusions and $P = P^\dagger$. So it is the rule $-i\hbar\psi'$ on a domain between $D(P_D)$ and $\mathcal{A}$. Two functions of $\mathcal{A}$ with the same values at the walls differ by an element of $D(P_D)$, so the domain is fixed by a condition on the boundary values. By (3.2) with $\varphi = \psi$, the mean $\langle \psi, -i\hbar\psi' \rangle$ is real only if $|\psi(L)| = |\psi(0)|$.

Definition 3.9 (The momenta P_θ). For $\theta \in [0, 2\pi)$, the momentum P_θ has the domain

$$D(P_\theta) = \{\psi \in \mathcal{A} : \ \psi(L) = e^{i\theta}\psi(0)\}$$

and the action $P_\theta\psi = -i\hbar\psi'$.

Theorem 3.10 (Self-adjoint momenta in a box). *For each $\theta \in [0, 2\pi)$, the momentum P_θ is self-adjoint, and $P_D \subseteq P_\theta \subseteq P_D^\dagger$. The functions*

$$e_{\theta,k}(x) = \frac{e^{i(\theta+2\pi k)x/L}}{\sqrt{L}} \qquad (k \in \mathbb{Z})$$

form a complete orthonormal basis of eigenvectors of P_θ, with the eigenvalues $\hbar(\theta + 2\pi k)/L$. The momentum P_0 is the circle's P.

Proof. First, $\psi(0) = \psi(L) = 0$ satisfies the condition, so $P_D \subseteq P_\theta$, and $P_\theta \subseteq P_D^\dagger$ by Proposition 3.7. For $\varphi, \psi \in D(P_\theta)$, $\overline{\varphi(L)}\psi(L) = \overline{e^{i\theta}\varphi(0)}\, e^{i\theta}\psi(0) = \overline{\varphi(0)}\psi(0)$, so (3.2) vanishes and P_θ is symmetric. Conversely, let $\varphi \in D(P_\theta^\dagger)$. Since $P_D \subseteq P_\theta$, the adjoint $P_\theta^\dagger$ is contained in $P_D^\dagger$, so by Theorem 3.8, $\varphi \in \mathcal{A}$ and $P_\theta^\dagger \varphi = -i\hbar\varphi'$. With $\psi = e_{\theta,0}$, (3.2) gives

$$0 = \langle \varphi, P_\theta \psi\rangle - \left\langle P_\theta^\dagger \varphi, \psi\right\rangle = -\frac{i\hbar}{\sqrt{L}}\left(\overline{\varphi(L)}e^{i\theta} - \overline{\varphi(0)}\right),$$

so $\varphi(L) = e^{i\theta}\varphi(0)$ and $\varphi \in D(P_\theta)$. Next, $e_{\theta,k} = e^{i\theta x/L} f_k$, and multiplication by $e^{i\theta x/L}$, of modulus 1, is unitary, so the $e_{\theta,k}$ form a complete orthonormal basis. Each lies in $D(P_\theta)$, and $-i\hbar e'_{\theta,k} = \hbar(\theta + 2\pi k)e_{\theta,k}/L$. Finally, $e_{0,k} = f_k$, and P_0 is a self-adjoint operator with $P_0 f_k = p_k f_k$. The argument of the step "the only one" of the proof of Theorem 1.8, with f_k and p_k, shows that $P_0 = P$. ∎

There are no other self-adjoint extensions of P_D. We also call these extensions the self-adjoint *versions* of the momentum.

Proposition 3.11 (These are all)**.** *Every self-adjoint operator P with $P_D \subseteq P$ is one of the P_θ.*

Proof. As shown before Definition 3.9, $P \subseteq P_D^\dagger$, so by Theorem 3.8 P is the rule $-i\hbar\psi'$ on a subspace $D(P)$ of $\mathcal{A}$ that contains $D(P_D)$. Let $V \subseteq \mathbb{C}^2$ be the set of pairs $(\psi(0), \psi(L))$ with $\psi \in D(P)$, a subspace. If $\psi \in \mathcal{A}$ has its pair in V, some $\varphi \in D(P)$ has the same pair, and $\psi - \varphi \in D(P_D) \subseteq D(P)$. So $D(P)$ is the set of $\psi \in \mathcal{A}$ whose pair lies in V. Since P is symmetric, every pair (a, b) in V has $|b| = |a|$, so $V \neq \mathbb{C}^2$. If $V = \{0\}$, then $P = P_D$, which is not self-adjoint (Proposition 3.7). So V is a line, spanned by a pair (a, b) with $|b| = |a| \neq 0$, that is, $b = e^{i\theta}a$. Then $D(P) = D(P_\theta)$, and $P = P_\theta$. ∎

Different θ give different eigenvalues, so the formula $-i\hbar\,\mathrm{d}/\mathrm{d}x$ does not decide the momentum. What does the choice change for a state?

Example 3.12 (One state, two momentum distributions)**.** Take the ground state u_1 of the box. It lies in $D(P_D)$, so in every $D(P_\theta)$, and every P_θ acts on it by $-i\hbar u_1'$.

1. *The rule.* By the observables postulate (Postulate 1.5) and Theorem 3.10, a reading of P_θ gives $\hbar(\theta + 2\pi k)/L$ with the probability $\left|\langle e_{\theta,k}, u_1\rangle\right|^2$.
2. *Read P_0.* Writing the sine through exponentials and integrating,

$$\langle f_k, u_1\rangle = \frac{\sqrt{2}}{L}\int_0^L \sin\frac{\pi x}{L}\, e^{-2\pi i k x/L}\,\mathrm{d}x = \frac{2\sqrt{2}}{\pi(1-4k^2)}.$$

   So P_0 reads 0 with probability $8/\pi^2 \approx 0.811$, each of $\pm 2\pi\hbar/L$ with probability $8/9\pi^2 \approx 0.090$, each of $\pm 4\pi\hbar/L$ with probability $8/225\pi^2 \approx 0.004$, and so on.
3. *Read P_π.* Since $e_{\pi,0} = e^{i\pi x/L}/\sqrt{L}$ and $e_{\pi,-1} = e^{-i\pi x/L}/\sqrt{L}$,

$$u_1 = \frac{e_{\pi,0} - e_{\pi,-1}}{i\sqrt{2}}.$$

   So P_π reads $\pi\hbar/L$ or $-\pi\hbar/L$, each with probability $\frac{1}{2}$.
4. *Compare.* For $\psi \in D(P_\theta)$, symmetry gives $\left\langle e_{\theta,k}, P_\theta\psi\right\rangle = \hbar(\theta + 2\pi k)c_k/L$, with $c_k = \left\langle e_{\theta,k}, \psi\right\rangle$. By Parseval's identity, the mean of the readings is $\langle\psi, P_\theta\psi\rangle$ and their mean square is $\|P_\theta\psi\|^2$. For u_1 these are 0 and $\hbar^2\left\|u_1'\right\|^2 = \pi^2\hbar^2/L^2$ for every θ. The two distributions have the same mean and the same mean square, and they are different.

The two lists sample one function. By Theorem 3.10, the coefficient $\langle e_{\theta,k}, \psi \rangle$ is the value at $p = \hbar(\theta + 2\pi k)/L$ of one function of p,

$$L^{-1/2} \int_0^L e^{-\mathrm{i}px/\hbar} \psi(x)\, \mathrm{d}x.$$

So every P_θ samples the squared modulus of this one function, and θ only shifts the grid of p. For $\psi = u_1$ and $\xi = pL/\hbar$, the squared modulus is

$$y(\xi) = \frac{8\pi^2 \cos^2(\xi/2)}{(\pi^2 - \xi^2)^2}, \qquad y(\pm\pi) = \tfrac{1}{2}.$$

Figure 3.2 shows it with the two grids of Example 3.12. Lecture 9 shows what θ does: it is the rule by which a translation crosses the walls.

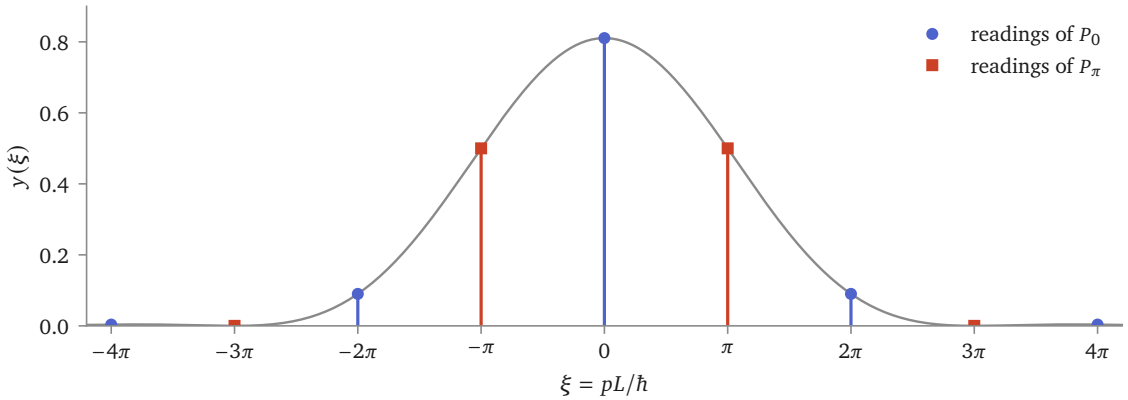


Figure 3.2: The ground state u_1 of the box. The curve is $y(\xi)$. Each stem is a probability of Example 3.12: blue at the readings $2\pi k\hbar/L$ of P_0, red at the readings $(2k+1)\pi\hbar/L$ of P_π. The stems are probabilities at single points, and the area under the curve is not a probability.

A ring with a magnetic flux realises P_θ. Put a flux through the ring of *Quantum Mechanics I*, §8.6: a hop to the next site gains the phase $e^{\mathrm{i}\beta}$, so $H_\beta = 2\kappa\mathbb{1} - \kappa(e^{\mathrm{i}\beta}T + e^{-\mathrm{i}\beta}T^\dagger)$. That lecture wrote θ for β, a letter that names the boundary condition here. Change the phases of the sites: write $\psi_x = e^{\mathrm{i}\beta x}\varphi_x$ for $x = 0, \ldots, N-1$. On the amplitudes φ_x, the hops inside the ring lose their phase, and the hop across from $x = N-1$ to $x = 0$ keeps the whole loop phase $e^{\mathrm{i}N\beta}$. In the notation of the chain, H_β becomes the hopping with $\varphi_N = e^{-\mathrm{i}N\beta}\varphi_0$, which is the condition of P_θ for $\theta \equiv -N\beta$, modulo 2π. Read from the right end to the left, the same condition is $\varphi_0 = e^{\mathrm{i}N\beta}\varphi_N = e^{-\mathrm{i}\theta}\varphi_N$. So the hop across, which also runs from the right end to the left, carries $e^{-\mathrm{i}\theta}$. Figure 3.3 draws the three steps.

The levels agree: with the loop phase $N\beta$ held fixed as $N \to \infty$,

$$4\kappa \sin^2 \frac{2\pi k - N\beta}{2N} \longrightarrow \frac{\hbar^2(2\pi k - N\beta)^2}{2mL^2}.$$

As k runs through $\mathbb{Z}$, these limits are the numbers $(\hbar(\theta + 2\pi k)/L)^2/2m$, $k \in \mathbb{Z}$, the levels of $P_\theta^2/2m$, after the integers are relabelled. The levels alone cannot fix the sign of θ. The phases of the sites do. In this sense the ring with a flux realises P_θ: the change of phases is exact on the ring, and the levels converge. As for the chain of § 1.1, this is not a limit of operators.

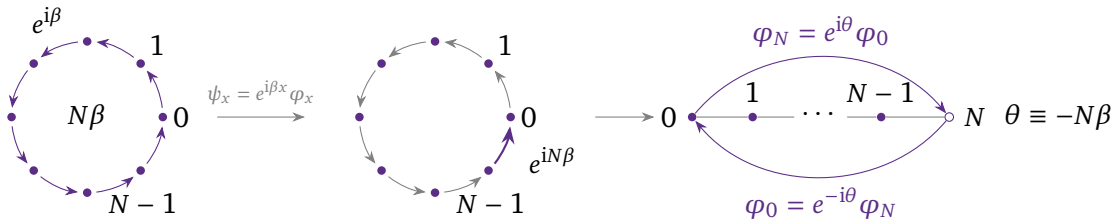


Figure 3.3: Left: the ring, drawn with 8 sites; each forward hop gains the phase $e^{i\beta}$, and a trip round the ring gains $e^{iN\beta}$. Middle: after the change of phases the interior hops carry no phase, and the hop from $x = N-1$ to $x = 0$ carries all of it. Right: the ring cut at that hop, with a site N at $x = L$ standing for the site 0. The boundary condition and its reverse have inverse phases.

On the ring with a flux, the flux fixes θ, and the energy $P_\theta^2/2m$ depends on it: there different θ are different systems. In the box the energy H is fixed (Theorem 1.8), and the P_θ are different observables of one system. Nothing in the box picks one. Choosing θ is a further part of specifying the reading. This lecture builds no apparatus that makes such a reading with the walls in place.

Self-check 3.4

1. Find the eigenvalues of P_π. For which θ is 0 an eigenvalue of P_θ?

The physics

A magnetic flux Φ through a ring changes the phase of a particle of charge q that goes once around it, in the direction of the forward hops, by $q\Phi/\hbar$, although the field is zero on the ring itself (the Aharonov–Bohm effect). For the charge $\pm e$ of an electron or a proton this phase is $\pm 2\pi\Phi/\Phi_0$, with the flux quantum $\Phi_0 = h/e$. So the flux sets the loop phase $N\beta = q\Phi/\hbar$, and with it the boundary condition. Since the levels depend on the flux, a ring of metal about a micrometre across, cooled to a fraction of a kelvin, can carry a current in equilibrium, with no battery. Lévy, Dolan, Dunsmuir and Bouchiat detected such persistent currents in copper rings in 1990, and Bleszynski-Jayich and coworkers measured them in aluminium rings in 2009.

3.5 The momentum and the box's energy

For a free particle, the energy is $P^2/2m$. In the box no self-adjoint momentum squares to the energy.

Proposition 3.13 (No momentum squares to the energy). *For no θ is $P_\theta P_\theta/2m$, on the $\psi \in D(P_\theta)$ with $P_\theta\psi \in D(P_\theta)$, the box Hamiltonian H.*

Proof. The function $e_{\theta,0}$ lies in $D(P_\theta)$, and so does $P_\theta e_{\theta,0} = (\hbar\theta/L)e_{\theta,0}$. So $e_{\theta,0}$ is an eigenvector of $P_\theta P_\theta$. If $P_\theta P_\theta/2m$ were H, then by Theorem 1.8 $e_{\theta,0}$ would be a multiple of some u_n. Both functions are continuous, and $u_n(0) = 0$, while $|e_{\theta,0}(0)| = 1/\sqrt{L}$. ∎

With the classification (Proposition 3.11), no self-adjoint version of the momentum squares to H. One might instead take a square root of $2mH$ as the momentum. The diagonal operator B with $Bu_n = (n\pi\hbar/L)u_n$, on the vectors with $\sum_n n^2 |c_n|^2 < \infty$, does satisfy $B^2/2m = H$, since $E_n = (n\pi\hbar/L)^2/2m$. But B is not a version of the derivative: $-i\hbar u_1'$ is a multiple of $\cos(\pi x/L)$,

not of u_1. The box energy is built from the momentum with walls in another way. The product $P_D^\dagger P_D$ acts on the $\psi \in D(P_D)$ with $P_D\psi \in D(P_D^\dagger) = \mathcal{A}$. On the twice continuously differentiable functions that vanish at the walls, the product divided by $2m$ is $(-\mathrm{i}\hbar)^2\psi''/2m = -(\hbar^2/2m)\psi''$. The next theorem describes the domain of H by derivatives, and this product turns out to be $2mH$.

Theorem 3.14 (The Dirichlet domain)**.** *The domain of the box Hamiltonian is*

$$D(H) = \{\psi \in \mathcal{A} : \ \psi' \in \mathcal{A},\ \psi(0) = \psi(L) = 0\},$$

and $H\psi = -(\hbar^2/2m)\,\psi''$ for every ψ in it.

Proof. We prove both inclusions. First, let $\psi \in \mathcal{A}$ with $\psi' \in \mathcal{A}$ and $\psi(0) = \psi(L) = 0$, and write $c_n = \langle u_n, \psi\rangle$. Integrate by parts twice with (3.1). The boundary terms vanish, since u_n and ψ vanish at the walls, and $u_n'' = -(n\pi/L)^2 u_n$, so

$$\langle u_n, -\psi''\rangle = \langle u_n', \psi'\rangle = -\langle u_n'', \psi\rangle = \left(\frac{n\pi}{L}\right)^2 c_n.$$

By Parseval's identity, $\sum_n (n\pi/L)^4 |c_n|^2 = \|\psi''\|^2 < \infty$, so $\psi \in D(H)$. Moreover, $H\psi = \sum_n E_n c_n u_n = (\hbar^2/2m)\sum_n (n\pi/L)^2 c_n u_n = -(\hbar^2/2m)\psi''$.

Conversely, let $\psi \in D(H)$, with $c_n = \langle u_n, \psi\rangle$, and put $h = \sum_n (n\pi/L)^2 c_n u_n$. This series converges in $L^2(0, L)$, since $\sum_n n^4 |c_n|^2 < \infty$. Let

$$\Phi(x) = \gamma x - \int_0^x \int_0^t h(r)\,\mathrm{d}r\,\mathrm{d}t,$$

with the number γ fixed by $\Phi(L) = 0$. Then $\Phi \in \mathcal{A}$, $\Phi' \in \mathcal{A}$, $\Phi'' = -h$ and $\Phi(0) = \Phi(L) = 0$. The first step, applied to Φ, gives $(n\pi/L)^2 \langle u_n, \Phi\rangle = \langle u_n, -\Phi''\rangle = \langle u_n, h\rangle = (n\pi/L)^2 c_n$, so $\langle u_n, \Phi\rangle = c_n$ for every n. By the sine basis theorem, $\psi = \Phi$. So ψ lies in the set of the theorem, and the first step gives its action. ∎

In particular, since $\psi' \in \mathcal{A}$, every vector of $D(H)$ has a continuous derivative on $[0, L]$.

Since $D(P_D^\dagger) = \mathcal{A}$ (Theorem 3.8), the domain above is exactly the domain of the product $P_D^\dagger P_D$, and on it $P_D^\dagger P_D\psi = (-\mathrm{i}\hbar)(-\mathrm{i}\hbar\psi')' = -\hbar^2\psi''$. So $H = P_D^\dagger P_D/2m$.

The Dirichlet condition $\psi(0) = \psi(L) = 0$ makes sense because a function of $\mathcal{A}$ has a value at every point. The tent τ lies in $D(P_D)$, but its derivative jumps, so $\tau' \notin \mathcal{A}$ and $\tau \notin D(H)$.

The diagonal operator B above has the domain $\sum_n n^2 |c_n|^2 < \infty$. Its condition is that of a finite mean energy, $\sum_n E_n |c_n|^2 < \infty$, and the momentum with walls describes it.

Proposition 3.15 (Finite mean energy)**.** *Let $\psi \in L^2(0, L)$, with $c_n = \langle u_n, \psi\rangle$. Then $\sum_n E_n |c_n|^2 < \infty$ if and only if $\psi \in D(P_D)$, and then*

$$\sum_{n=1}^{\infty} E_n |c_n|^2 = \frac{\|P_D\psi\|^2}{2m} = \frac{\hbar^2}{2m}\int_0^L |\psi'(x)|^2\,\mathrm{d}x.$$

Proof. The cosines $v_n(x) = \sqrt{2/L}\,\cos(n\pi x/L)$, $n \geq 1$, are orthonormal, by the integral of a product of two cosines, and $v_n' = -(n\pi/L)u_n$. First let $\psi \in D(P_D)$. By integration by parts (3.1), with ψ vanishing at the walls,

$$\langle v_n, \psi'\rangle = -\langle v_n', \psi\rangle = \frac{n\pi}{L}\,c_n.$$

By Bessel's inequality, $\sum_n (n\pi/L)^2 |c_n|^2 \le \|\psi'\|^2$, and $E_n = (\hbar^2/2m)(n\pi/L)^2$. So the mean energy is finite. Conversely, let $\sum_n n^2 |c_n|^2 < \infty$. Then $g = \sum_n (n\pi/L) c_n v_n$ converges, with $\|g\|^2 = \sum_n (n\pi/L)^2 |c_n|^2$. Put $\varphi(x) = \int_0^x g(t)\,dt$. Then $\varphi \in \mathcal{A}$, $\varphi(0) = 0$, and $\varphi(L) = \langle 1, g\rangle = 0$, since each v_n has the integral 0. So $\varphi \in D(P_D)$, and by the first step $\langle u_n, \varphi\rangle = (L/n\pi)\,\langle v_n, g\rangle = c_n$ for every n. By the sine basis theorem $\varphi = \psi$, so $\psi \in D(P_D)$, with $\psi' = g$ by the second fact about $\mathcal{A}$. Hence $\|\psi'\|^2 = \sum_n (n\pi/L)^2 |c_n|^2$, and multiplying by $\hbar^2/2m$ gives the formula. ∎

So the state of § 1.7, with $c_n \propto 1/n^2$, lies in $D(P_D)$ but not in $D(H)$. The state of § 2.6 that a reading of the left half leaves jumps at $L/2$, so it is not in $\mathcal{A}$, and its mean energy is infinite, as found there. The operators B and P_D have the same domain, and $\|B\psi\| = \|P_D\psi\|$ on it, but they are different operators.

The domains of the box are nested. The finite sums of the u_n lie in $D(H)$, which holds the ψ with $\sum_n E_n^2 |c_n|^2 < \infty$. Since $E_n = n^2 E_1$ and $n^2 \le n^4$, every vector of $D(H)$ lies in $D(P_D)$, which holds the ψ with $\sum_n E_n |c_n|^2 < \infty$ by Proposition 3.15. Figure 3.4 draws these sets inside $L^2(0, L)$ and places six vectors of Lectures 1 to 3 in them.

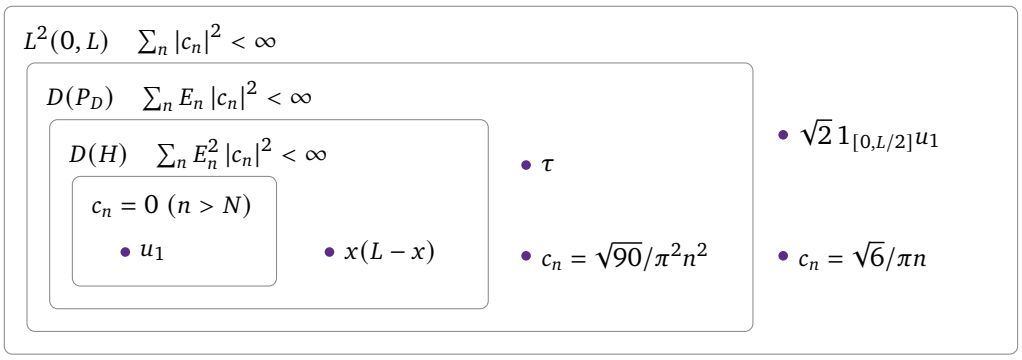


Figure 3.4: The sets of vectors of the box, nested from the finite sums of the u_n to $L^2(0, L)$. Each set carries its name and a condition on the coefficients $c_n = \langle u_n, \psi\rangle$ that describes it (for $D(P_D)$, by Proposition 3.15). For the finite sums, N depends on ψ. Each dot is a vector, drawn in the smallest set that contains it. The vector $x(L-x)$ is the parabola of Example 1.9 up to normalisation, $c_n = \sqrt{90}/\pi^2 n^2$ is the state of § 1.7, and $\sqrt{2}\,1_{[0,L/2]}u_1$ is the state of § 2.6.

Self-check 3.5

1. Does u_1 lie in $D(P_\theta)$ for every θ? Does $P_\theta u_1$ depend on θ? For which θ is $P_\theta u_1$ in $D(P_\theta)$, and what is $P_\theta P_\theta u_1$ then?

Momentum on a circle and in a box

On the circle: the momentum P, diagonal in the Fourier basis with the eigenvalues $2\pi\hbar k/L$, is self-adjoint, and the energy has the levels $p_k^2/2m$, the limits of the ring's levels (Proposition 3.2, § 3.1). *In the box:* the momentum with walls P_D is symmetric and not self-adjoint (Propositions 3.6 and 3.7). Its self-adjoint extensions are the P_θ, $\psi(L) = e^{i\theta}\psi(0)$, with the eigenvalues $\hbar(\theta + 2\pi k)/L$, and none squares to the box's energy (Theorem 3.10 and Propositions 3.11 and 3.13). They are different observables of the box, with different readings on one state (Example 3.12). The ring with a flux realises them in the sense of § 3.4, and there θ fixes the system. *Energy:* the mean energy is finite exactly on $D(P_D)$ (Proposition 3.15).

Summary.

- *The momentum on a circle* is self-adjoint (Proposition 3.2), and the ring's levels tend to the circle's (§ 3.1).
- *The momentum with walls* P_D is symmetric, its adjoint has every complex number as an eigenvalue, and it is not self-adjoint (Propositions 3.6 and 3.7). Its adjoint is the rule on all of $\mathcal{A}$ (Theorem 3.8).
- *Self-adjoint momenta in a box*: each P_θ is self-adjoint, with the eigenbasis $e_{\theta,k}$, and every self-adjoint extension of P_D is one of them (Theorem 3.10 and Proposition 3.11). A flux β per hop on the ring gives the boundary condition of P_θ with $\theta \equiv -N\beta$ (§ 3.4).
- *One state, two momentum distributions*: u_1 reads 0 under P_0 with probability $8/\pi^2$, and $\pm\pi\hbar/L$ under P_π with probability $\frac{1}{2}$ each (Example 3.12).
- *The Dirichlet domain*: $D(H)$ holds the $\psi \in \mathcal{A}$ with $\psi' \in \mathcal{A}$ and $\psi(0) = \psi(L) = 0$, and $H = P_D^\dagger P_D/2m$ (Theorem 3.14).
- *Finite mean energy* is the domain of the momentum with walls, and the mean energy is $(\hbar^2/2m)\int |\psi'|^2$ (Proposition 3.15).
- *No momentum squares to the box's energy* (Proposition 3.13).

Lecture 4 lets the circle grow into the line, where position and momentum have no eigenvalues at all, and proves the uncertainty relation there.

Self-check answers.

3.1 1. Circle: 0 once, $4E_1$ twice, $16E_1$ twice. Box: E_1, $4E_1$, $9E_1$, each once.

3.2 1. Yes: it is continuously differentiable and vanishes at both walls. $\langle\psi, P_D\psi\rangle = -\mathrm{i}\hbar\int_0^L x(L-x)(L-2x)\,\mathrm{d}x = 0$. 2. The indicator jumps, so it is not in $\mathcal{A}$. The function $x^{2/3}$ is, with $g = \frac{2}{3}x^{-1/3}$ and $\int_0^L g^2\,\mathrm{d}x = \frac{4}{3}L^{1/3}$; it is not in $D(P_D)$, since it is not 0 at L. By the product rule, $x^{2/3}(L-x)$ lies in $\mathcal{A}$. It vanishes at both walls, so it lies in $D(P_D)$.

3.3 1. $e^{-x/L}$, since $-\mathrm{i}\hbar(e^{-x/L})' = (\mathrm{i}\hbar/L)e^{-x/L}$. Its value at 0 is 1, so it is not in $D(P_D)$.

3.4 1. $(2k+1)\pi\hbar/L$ for $k \in \mathbb{Z}$, symmetric about 0. Only for $\theta = 0$.

3.5 1. Yes, since $u_1(0) = u_1(L) = 0$. No: every P_θ acts on it by $-\mathrm{i}\hbar u_1'$, a multiple of $\cos(\pi x/L)$, whose value at L is -1 times its value at 0. So $P_\theta u_1 \in D(P_\theta)$ only for $\theta = \pi$, and then $P_\pi P_\pi u_1 = -\hbar^2 u_1'' = 2mE_1 u_1$.

References

Teschl, *Mathematical Methods in Quantum Mechanics*, (2.28)–(2.37) (pp. 61–62), (2.50)–(2.57) (pp. 69–70), Thm 2.26 and (2.110) (pp. 82–83). Teta, *A Mathematical Primer on Quantum Mechanics*, Ex. 4.22 (pp. 101–102). Amrein, *Hilbert Space Methods in Quantum Mechanics*, (3.22) (p. 103) and (3.29) (p. 109). *Quantum Mechanics I*, Lecture 8. *Linear Algebra for Physics*, Ex. 13.25. Lévy, Dolan, Dunsmuir and Bouchiat, *Physical Review Letters* 64, 2074 (1990). Bleszynski-Jayich et al., *Science* 326, 272 (2009).

Teschl, (2.32), gives the same adjoint as Theorem 3.8 for $-\mathrm{i}\,\mathrm{d}/\mathrm{d}x$ on the continuously differentiable functions on $[0, 2\pi]$ that vanish at both ends. Amrein, (3.22), and Teschl, Thm 2.26 and (2.110), whose $\tilde{\theta}$ is our θ, give the same family as Theorem 3.10 and Proposition 3.11. Amrein, (3.29), states the identity $H = P_D^\dagger P_D/2m$. Teschl, (2.57), gives the domain of Theorem 3.14, and Teta, Ex. 4.22, proves the uniqueness of H by another route. Both take $\hbar = 1$, $m = \frac{1}{2}$ and $L = \pi$.